

DBMS project proposal

Yassin Ahmed 202300304
Abdelrahman Adel 202301698
Abdelrahman Taha 202300062

Problem: Clinic web app

As someone from Gen-Z, who has a daily average screen-time of 6 hours+, it would be very convenient to have an application which gives me a list of doctors nearby that I can contact. Moreover, if i could book doctor appointments based on others' ratings. Thus, we have come up with this project idea where doctors and patients can come into contact with each other.

Users: Privileges and Obligations

Patient

- To register as a patient, a National ID must be provided, it is not urgent to validate this however.
- The patient can add X-Rays/MRI's and their medical history after registration
- Can view all doctors currently available and all their details by searching via the doctor's name.
- Can book appointments at any time slot available in the doctors' schedules.
 - Afterwards, an email gets sent to the Doctor to confirm the appointment. The patient gets notified whenever the doctor accepts or rejects the request.
- Can view doctor's notes for diagnosis after appointment is done when the doctor adds them.
- Can rate and add comments on their experience with the doctor after an appointment.
- Can view all previously booked appointments and notes written there. Eventually, a "portfolio" or history of the patient should be built based on the notes and medical history (maybe use chatgpt's api? ;))

Doctors

- Must provide national ID
- Must provide authentic certificates showing that he has studied medicine
- Must provide work experience and papers that validate his work in hospitals/clinics.

- Can view all available patients with current conditions that match the doctor's expertise / specialization.
- Name their price per appointment
- Accepts/Rejects appointment requests from patients.
- Can add diagnosis notes for patients after appointment

Admins

- Validates National ID's for doctors and patients.
- Validates doctors' certificates.
- Can ban doctors/patients at will but with valid reason.
- Must accept all doctors and patients before getting to use the app.
- View statistics on the (newly registered patients) / (number of booked appointments) / (Most/least booked doctors) per month/week/day.
- Have the viewing privileges of both doctors and patients (they can see a table of patients and doctors, and navigate them through a search via national id or name or even field of medicine for the doctors).

Its important to note that we plan on making "Admins" an automated procedure in the future, currently they're just moderators and are necessary for the safety of the users.

Entities and attributes

User

- National ID
- Registration date
- Sex
- Birthdate
- Age (derived)
- Location
- Name
- Username (unique primary key)
- Password

Doctors

- Certificates (entity)
- Schedule
- Price per appointment
- Work experience (entity)
- Field of medicine (entity)
- Location of current work

Patients

- Conditions (entity)

Admins

- Has an account
- Ban privileges
- Approval privileges

Conditions (weak entity)

- Current conditions
 - Condition (multivalued) (entity)
- Previous conditions
 - Condition (multivalued) (entity)
- ID of owner

Condition (weak entity)

- Description
- How it was treated
- Date and time of identification
- Lifespan

Certificates (weak entity)

- Certificate (multivalued)
 - Description
 - Date
 - Where from

Work experience

- Experience (multivalued)
 - Location/Institution
 - Span (number of years/months)
 - Position
 - Proof (did not decide how yet we can validate this)

Field of medicine

- Name
- Average price (derived from the doctors who are in a relationship with this field)

Relationships

Admin validations

- Admin validates National id of doctors
- Admin validates National id of patients

Admin punishments

- Ban doctors with valid reason.
- Add fees to patients if they book appointments and cancel them more than 3 times.

Doctor papers

- Doctor can add work experience to experience entity
- Can add certificates

Appointments

- Doctors can add N time slots which they are available at.
- Patients can book multiple appointments with multiple time-slots.
- Doctors can confirm said appointments so they are finalized and the patient knows that the doctor has been successfully notified.
- Doctors, patients and their appointments share a ternary relationship where doctors write down his/her diagnosis after the appointment has taken place. This "writing down" happens inside the appointment entity in the "diagnosis" attribute.

Conditions

- A patient can have N conditions
- Said condition must take a single value of a table of prior-known conditions.

Doctor specialization

Doctors can specialize in one field of medicine from a table of already pre-made fields.

Patient review

Patients can review doctors after appointments have finished, which leaves behind 2 attributes: comments (can be null) and the number of stars given to the doctor.